3.4.3 Database Design

FitnessBalance uses Cloud Firestore as its principal cloud database. Shared catalogue collections store food, exercise and reward definitions, while private logs, preferences and progress are organised beneath the authenticated user's profile document. Document-oriented modelling supports grouping related information around application access patterns (Sadalage & Fowler, 2012). In FitnessBalance, the Firebase Authentication UID identifies the users document and determines ownership of its private subcollections. Passwords are managed by Firebase Authentication and are not stored in Firestore.

This section summarises the main data structures and the reasons for their organisation. The complete field dictionaries, including data types, optional fields, embedded records and supporting receipts, are provided in Appendix A. The simplified ERD in Section 3.4.4 illustrates the core relationships; it does not imply that the supporting collections have been removed.

3.4.3.1 User Information and Preferences

The users collection stores personal profile information, explicit nutrition goals and workout preferences. Preference fields support the implemented food and workout recommendation rules, while protected progress fields provide the user's accumulated XP and level. Some fields are created only after preferences are configured or an eligible workout is recorded. Table 3.3 groups the main fields; Appendix A.1 provides the detailed dictionary.

Table 3.3: User information and preference groups

Data group	Main fields	Purpose
Profile	fullName, email, weight, height, fitnessLevel	Stores personal information and the current profile measurements.
Nutrition goals	dailyCalorieGoal, dailyProteinGoalG, dailyCarbohydrateGoalG, dailyFatGoalG	Stores explicitly selected calorie and macronutrient targets.
Workout preferences	preferredWorkoutMinutes, preferredWorkoutDays, weeklyWorkoutTarget, equipment	Supports workout scheduling, eligibility and duration preferences.
Food preferences	dietaryPreferences, preferredFoodIds, dislikedFoodIds	Supports filtering and ranking of eligible food suggestions.
Progress summary	totalXp, level, achievementIds, metGoalWeeks	Provides accumulated application progress and achieved target weeks.

3.4.3.2 Food and Nutrition Data

The foods collection contains the Malaysian food catalogue imported into Firestore. Each definition preserves available source nutrition, its reference basis and provenance. Food consumption is stored separately in dated foodLogs documents as embedded foodItems snapshots. These snapshots retain the

selected portion and the available nutrition at logging time, so later catalogue changes do not silently rewrite historical consumption.

Saved meals and recipes reuse ingredient or portion snapshots, while personal servings provide user-defined labels tied to measurable amounts. Private barcode products and personal nutrition overrides are stored under the user's account rather than modifying the shared catalogue. Missing values remain distinguishable from zero, and aggregate totals retain coverage information. Table 3.4 summarises the structures; Appendix A.2–A.5 and A.15–A.19 provide their detailed dictionaries.

Table 3.4: Food and nutrition data structures

Collection / subcollection	Main fields	Purpose
foods	food_name, nutrition_basis, nutrition, source_database	Stores shared food definitions and source reference nutrition.
foodLogs	date, foodItems, totalCalories, nutrientTotals, nutrientCoverage	Stores dated consumption snapshots and known nutrient totals.
savedMeals	name, items, revision	Stores reusable groups of food portions.
recipes	name, items, yieldServings, revision	Stores ingredient snapshots and recipe yield.
personalServings	foodKey, name, amount, unit	Defines a private serving label with a measurable amount.
barcodeFoods	barcode, food_name, nutrition, label_nutrition	Stores user-entered private product definitions.
foodNutrition	originalFoodId, nutrition, source_kind, revision	Stores private label-based or approximate alternative nutrition.

3.4.3.3 Exercise Data

Exercise content is separated from user activity. Shared exercises define reusable movements and instructions. workoutPaths contain lessons, and each lesson contains ordered workoutExercises assignments referencing the exercise definitions. This allows the same exercise to be used in different lessons without duplicating its instructional content.

A guided workout creates a workoutSessions start record and, upon completion, an immutable workoutRecords result using the same session identifier. The result preserves completion, duration, estimated calories, feedback and awarded progress. Manual activities and explicitly confirmed wearable sessions use the separate dated exerciseLogs structure. Table 3.5 summarises these records; Appendix A.7–A.14 and A.22 provide the field dictionaries.

Table 3.5: Exercise catalogue and activity structures

Collection / subcollection	Main fields	Purpose
exercises	name, instructions, targetType, equipment, metValue	Defines reusable exercises and estimation inputs.

Collection / subcollection	Main fields	Purpose
workoutPaths	name, description, active	Organises guided workout content into paths.
lessons	sequence, difficulty, estimatedMinutes, baseXp, prerequisiteLessonId	Defines lesson information and progression eligibility.
workoutExercises	exerciseId, sequence, targetType, targetValue	Assigns ordered exercise targets within a lesson.
workoutSessions	pathId, lessonId, plannedExercises, startedAt	Preserves the guided workout's initial context.
workoutRecords	lessonId, completionPercentage, durationSeconds, feedback fields, xpEarned	Stores immutable guided outcomes and reward results.
exerciseLogs	date, exercises, totalCaloriesBurned	Stores dated manual and confirmed wearable activity entries.
pathProgress	completionCounts, bestPercent, bestStars	Tracks per-user attempts and best lesson results.

3.4.3.4 Progress and Gamification Data

dailyData stores dated food and exercise summaries for the dashboard and progress views. This controlled denormalisation reduces the need to reconstruct every summary from its source records during each read, but requires coordinated updates to keep duplicated values consistent (Kleppmann, 2017). Food and exercise operations therefore update their relevant summary fields without replacing unrelated totals.

Dated weight measurements are stored in weightLogs, with weightHistory maintaining a private index for history operations. Gamification uses exerciseWeeks for shared weekly progress, questProgress for dated quest completions, and unlockedAchievements for earned achievement evidence. Shared quests and achievements provide definition metadata. The design tracks weekly targets and Momentum rather than consecutive-day streaks. Table 3.6 summarises these structures; Appendix A.6 and A.20–A.27 contain their detailed dictionaries.

Table 3.6: Progress and gamification structures

Collection / subcollection	Main fields	Purpose
dailyData	caloriesConsumed, caloriesBurned, nutrientTotals, exerciseEntryCount	Provides dated dashboard summaries and completeness information.
weightLogs	day, weightKg, revision	Stores manual weight measurements by date.
weightHistory	entries, updatedAt	Maintains the private weight-history index.
exerciseWeeks	completed, target, categories, bonusAwarded	Stores shared weekly workout progress and bonus status.

Collection / subcollection	Main fields	Purpose
quests	name, target, bonusXp	Stores shared quest definition metadata.
questProgress	day, questId, recordId, xpEarned	Records immutable daily quest completions.
achievements	name, description, active	Stores shared achievement display metadata.
unlockedAchievements	recordId, unlockedAt	Records the evidence for each earned achievement.

3.4.3.5 Health Data and Data Integrity

Health Connect synchronisation uses private, account-specific local caches rather than automatically creating cloud health-record collections. Summaries and a rolling wearable-session cache remain on the device until the user explicitly confirms a wearable workout. Confirmation saves the selected activity metadata, elapsed duration and available active calories into exerciseLogs. Qualifying sessions can also create separate reward evidence. Raw health samples and the raw energy-interval list are not uploaded by this workflow.

Supporting receipts prevent a repeated meal insertion or wearable confirmation from creating duplicate contributions. Wearable award and date-credit records additionally prevent repeated XP and weekly credit. Firestore transactions apply their writes atomically and can retry when concurrent changes affect their reads (Google, n.d.). FitnessBalance combines transactions, stable identifiers and revision checks to coordinate related log and progress changes. Table 3.7 summarises the cloud support records; their fields and the local-storage inventory are provided in Appendix A.28–A.32.

Firestore security rules restrict private records to the owner identified by the parent UID and protect shared catalogue data from ordinary client writes. Stored identifiers express logical associations rather than SQL foreign-key constraints. Supporting reward receipts are retained under the implemented policy even if a confirmed activity is later removed from calorie and duration totals. Appendix A.33 summarises the logical collection relationships.

Table 3.7: Supporting confirmation and consistency records

Supporting subcollection	Main fields	Purpose
savedMealLogs	date, mealId, entryCount, createdAt	Prevents repeated insertion of the same reusable-meal operation.
wearableWorkoutLinks	date, entryId, createdAt	Identifies a wearable session already confirmed into the activity log.
wearableWorkoutRewards	entryId, day, week, xpEarned, ruleVersion	Preserves the immutable result of an eligible wearable award.
wearableRewardDays	day, week, awardId	Limits wearable base XP and weekly credit to the first qualifying award on that date.

Appendix A: Database Data Dictionary

This appendix provides the detailed structures summarised in Section 3.4.3. It contains 28 collection/subcollection document dictionaries and three additional dictionaries for embedded maps or array records. Embedded structures are not independent Firestore entities. Local storage and relationship summaries are included at the end.

Nullable = false means that a field is expected in the current creation workflow; Nullable = true means it may be absent, null or applicable only to a particular workflow. Firestore does not enforce relational NOT NULL constraints, and older documents can have fewer fields. Document IDs are explicitly identified as path keys rather than automatically duplicated stored fields. A numeric zero is distinct from an unavailable value. These dictionaries describe repository-defined structures, not a field-frequency audit of the live database.

A.1 Users collection

The users collection stores account profile information, explicit nutrition and workout preferences, and the user's accumulated application progress. Its document identifier is the Firebase Authentication UID. Passwords are managed by Firebase Authentication and are not stored in the user document.

Table A.1: Users collection

Field	Type	Nullable	Description
uid (document ID)	string	false	Firebase Authentication UID used as the document path identifier; not a separate stored field.
fullName	string	false	Name supplied during registration and updated through the profile screen.
email	string	false	Registered email address copied into the profile; authentication identity remains managed by Firebase Authentication.
weight	number	false	Current profile body weight in kilograms.
height	number	true	Profile height in centimetres, supplied after registration.
fitnessLevel	string	false	Selected fitness level used by workout preferences.
createdAt	timestamp	false	Account profile creation time.

Field	Type	Nullable	Description
isVerified	boolean	false	Profile metadata initially set to false; Firebase Auth emailVerified is the authoritative verification state.
nutritionGoalSource	string	false	Indicates whether nutrition goals are not set or explicitly user-entered.
dailyCalorieGoal	number	true	Explicit daily energy target in kilocalories; absent until configured.
dailyProteinGoalG	number	true	Explicit daily protein target in grams.
dailyCarbohydrateGoalG	number	true	Explicit daily carbohydrate target in grams.
dailyFatGoalG	number	true	Explicit daily fat target in grams.
preferredExercises	array(string)	true	Activity preferences selected through the profile workflow.
preferredWorkoutMinutes	integer	true	Preferred workout duration in minutes.
preferredWorkoutDays	array(integer)	true	Chosen workout weekdays, encoded from 1 (Monday) to 7 (Sunday).
weeklyWorkoutTarget	integer	true	Weekly target bounded from one to seven workouts.
equipment	array(string)	true	Available equipment identifiers, or none.
dietaryPreferences	array(string)	true	Supported dietary preferences used to restrict eligible food suggestions.
preferredFoodIds	array(string)	true	Food identifiers marked as preferred for recommendation ranking.

Field	Type	Nullable	Description
dislikedFoodIds	array(string)	true	Food identifiers excluded by the user's recommendation preferences.
preferencesUpdatedAt	timestamp	true	Time of the latest preferences update.
totalXp	integer	true	Accumulated XP, created or updated by an eligible reward transaction.
level	integer	true	Application level derived from total XP.
lastRecordId	string	true	Most recent guided record or wearable award associated with an atomic progress update.
metGoalWeeks	array(string)	true	Distinct Monday-based week identifiers for which the weekly target bonus was earned.
achievementIds	array(string)	true	Identifiers of achievements recorded in the user's progress summary.
feedbackCount	integer	true	Count of saved guided-workout feedback submissions.
lastWorkoutAt	timestamp	true	Most recent qualifying workout time used by progress and inactivity logic.
lastDiscomfortAt	timestamp	true	Latest recorded discomfort feedback time.
lastDifficulty	string	true	Latest guided-workout difficulty feedback.
lastFeeling	string	true	Latest guided-workout feeling feedback.
timeZone	string	true	Time-zone identifier recorded by guided reward processing.

Progress and preference fields may be absent on a newly registered account. Legacy weeklyCounts and weeklyCategories may still be read for compatibility; they are not the primary current week-document design. The recommendation engine can read fitnessGoal, but the current profile/preferences UI does not persist a named fitnessGoal selector, so it is not presented as a mandatory implemented profile field.

A.2 Foods collection

The foods collection contains the shared food catalogue imported from Malaysian food composition datasets. It separates source reference nutrition from the serving-adjusted snapshots saved in individual food logs. Additional reviewed metadata supports food recommendations without changing the original source nutrients.

Table A.2: Foods collection

Field	Type	Nullable	Description
foodKey (document ID)	string	false	Stable identifier of a shared catalogue document.
food_key	string	false	Imported stable food identifier corresponding to the document key.
food_id	string	true	Identifier assigned by the original source dataset.
food_name	string	false	Food name presented to the user.
food_name_lowercase	string	false	Normalised food name used by prefix search.
food_group	string	true	Source food-group code, where supplied.
source_database	string	false	Origin of the record, such as a MyFCD dataset.
nutrition_basis	string	false	Reference basis, such as per_100g, per_100ml or unknown.
portion_unit	string	true	Compatible portion measurement unit when established.
nutrition	map	false	Available reference nutrient values; individual nutrient keys can be absent.

Field	Type	Nullable	Description
additional_nutrients	map	true	Additional source nutrient measurements with their original names and units.
image_url	string	true	Image address when available; may be null.
source_url	string	true	Source-page URL, if retained in the imported record.
nutrition_verification	string	true	Provenance or verification description supplied by the data-processing workflow.
basis_confirmation_required	boolean	true	Indicates that a reference basis needs explicit confirmation before automatic scaling.
recommendation_ready	boolean	true	Explicit flag enabling consideration by the reviewed recommendation catalogue.
meal_types	array(string)	true	Reviewed meal categories for recommendation eligibility.
serving_options	array(map)	true	Reviewed portion options containing label, amount and unit.
dietary_tags	array(string)	true	Reviewed dietary classifications; unknown suitability is not inferred from the name.
dietary_tags_status	string	true	Review status associated with dietary classifications.
dietary_evidence	map	true	Evidence notes keyed by dietary classification.
recommendation_metadata_version	integer	true	Version of the reviewed recommendation metadata.

Imported documents can contain further source-specific top-level nutrient or provenance fields. This table describes the core application contract rather than asserting that every live document has been inspected. Recommendation fields are conditional and are not automatically populated for all imported foods.

A.3 Nutrition map

The nutrition map stores the available reference values for the source food. The same canonical nutrient keys are used in portion-adjusted food-entry nutrients and aggregate nutrient maps, with different reference contexts.

Table A.3: Nutrition map

Field	Type	Nullable	Description
energy_kcal	number	true	Energy in kilocalories for the stated reference basis.
protein_g	number	true	Protein in grams for the stated reference basis.
carbohydrate_g	number	true	Carbohydrate in grams for the stated reference basis.
fat_g	number	true	Fat in grams for the stated reference basis.
fiber_g	number	true	Dietary fibre in grams for the stated reference basis.
sodium_mg	number	true	Sodium in milligrams for the stated reference basis.

This is an embedded map, not a collection. Missing values remain unavailable. A recorded numeric zero is different from a missing measurement. Additional source-specific nutrition keys may be retained. Energy estimation/provenance metadata is documented separately when present; this table does not certify nutrient accuracy.

A.4 FoodLogs subcollection

The foodLogs subcollection groups food-consumption snapshots by local date. Each document contains the food entries for that day and recalculated totals with completeness information.

Table A.4: FoodLogs subcollection

Field	Type	Nullable	Description
date (document ID)	string	false	Date identifier in yyyy-MM-dd format.
date	string	false	Local log date stored in the document.

Field	Type	Nullable	Description
foodItems	array(map)	false	Food-entry snapshots logged for the selected date.
totalCalories	integer	false	Rounded sum of available energy values; not a complete intake estimate when entries lack calories.
nutrientTotals	map	false	Sums of available canonical nutrients from the day's entries.
nutrientCoverage	map	false	Number of entries contributing a known value to each nutrient total.
knownCalorieCount	integer	false	Number of entries with usable calorie information.
missingCalorieCount	integer	false	Number of entries without usable calorie information.
caloriesComplete	boolean	false	Indicates whether all logged entries have usable energy values.
timestamp	timestamp	false	Latest write time for the dated food log.

A.5 FoodItems embedded records

A foodItems element preserves the food name, chosen serving and nutrition values as they were recorded. This snapshot design prevents later catalogue changes from silently altering historical consumption.

Table A.5: FoodItems embedded records

Field	Type	Nullable	Description
entryId	string	false	Stable entry identifier used for editing, deletion and duplicate protection.
foodId	string	true	Source food identifier when one is available.
name	string	false	Food-name snapshot.

Field	Type	Nullable	Description
meal	string	false	Selected meal category: Breakfast, Lunch, Dinner or Snacks.
servingAmount	number	true	Amount in one serving, when specified quantitatively.
servings	number	true	Number of selected servings.
portionUnit	string	true	Portion unit such as g or ml.
totalAmount	number	true	Total quantity calculated from serving amount and count.
portionGrams	number	true	Total weight when the selected unit is grams.
portionMillilitres	number	true	Total volume when the selected unit is millilitres.
calories	number	true	Known/calculated calories for the consumed portion; absent or unavailable for unknown-energy entries.
nutrients	map	true	Available portion-adjusted nutrient values using canonical keys.
nutritionBasis	string	true	Reference basis used or retained when logging.
sourceDatabase	string	true	Source description copied into the historical entry.
barcode	string	true	Product barcode for barcode-based records.
sourceUrl	string	true	Source URL, where supplied.
sourceRetrievedAt	timestamp or string	true	Source retrieval metadata copied as supplied by the source record.

Field	Type	Nullable	Description
basisConfirmationSource	string	true	Describes source metadata, user-label entry or another supported reference confirmation.
calculationMode	string	true	Indicates the supported calculation path, such as database, manual or macro estimate.
energyMethod	string	true	Provenance of energy, such as source-reported, user-entered or Atwater estimate.
nutritionVerification	string	true	Describes whether values were user-entered, approximate or not independently verified.
manualCaloriesPerServing	number	true	Explicit calories entered for one serving in the manual workflow.
portionDescription	string	true	Descriptive portion text used by unknown-value logging.
nutritionStatus	string	true	Availability status for an incomplete entry.
savedMealId	string	true	Originating saved meal, where applicable.
savedMealName	string	true	Saved-meal name snapshot.
savedMealOperation	string	true	Operation identifier tying a multi-food insertion to its receipt.
recipeId	string	true	Originating recipe identifier.
recipeName	string	true	Recipe-name snapshot.
recipeRevision	integer	true	Recipe version used to generate the entry.
recipeYield	number	true	Yield of the recipe version used.
recipeServings	number	true	Number of recipe servings consumed.

Field	Type	Nullable	Description
personalNutritionKind	string	true	Kind of personal nutrition override applied.
personalNutritionRevision	integer	true	Version of the personal override used.
alternativeFoodId	string	true	Selected approximate alternative food, where applicable.
alternativeName	string	true	Alternative food-name snapshot.
loggedAt	timestamp	true	Timestamp added to reusable-meal/recipe insertions where recorded.

These records are embedded in an array and are not separate top-level MealRecords documents. Optional provenance fields depend on the logging path; older entries can have fewer fields.

A.6 DailyData subcollection

The dailyData subcollection provides dated dashboard summaries for food consumption and exercise. Food and exercise repositories update their respective fields using merges and transactions so that one module does not replace the other module's totals.

Table A.6: DailyData subcollection

Field	Type	Nullable	Description
date (document ID)	string	false	Local date represented by the summary document.
caloriesConsumed	integer	true	Rounded known food-energy total for the date.
nutrientTotals	map	true	Available food nutrient totals.
nutrientCoverage	map	true	Known-value coverage for each food nutrient.
foodEntryCount	integer	true	Number of food entries included.
knownCalorieCount	integer	true	Food entries with usable calories.
missingCalorieCount	integer	true	Food entries with unavailable calories.
caloriesComplete	boolean	true	Whether food-energy coverage is complete.

Field	Type	Nullable	Description
manualCaloriesBurned	integer	true	Exercise-energy contribution from the legacy activity-log bucket, including confirmed wearable entries.
structuredCaloriesBurned	integer	true	Contribution from guided workout records.
caloriesBurned	integer	true	Combined known exercise calories after source-aware accounting.
exerciseActiveSeconds	integer	true	Combined counted exercise time; manual/wearable durations can represent elapsed time despite the historical field name.
exerciseEntryCount	integer	true	Number of counted exercise contributions.
exerciseTotalsVersion	integer	true	Version of the exercise aggregation representation.
exerciseCaloriesUnavailableCount	integer	true	Counted activity entries whose calories are unavailable.
structuredExerciseContributions	map	true	Guided contributions keyed by workout record ID, including any duplicate-suppression metadata.
lastGuidedCompletedAt	integer	true	Epoch milliseconds of the latest guided completion recorded by the transaction.
lastUpdated	timestamp	false	Latest summary update time.

Food-only and exercise-only days can contain different subsets of these fields. The document ID supplies the date; current writers do not require a separate date field here. This summary is intentionally denormalised for read efficiency.

A.7 Exercises collection

The exercises collection stores reusable exercise definitions for guided workout lessons. It contains instructional content, targets and parameters used by the existing exercise calorie-estimation logic.

Table A.7: Exercises collection

Field	Type	Nullable	Description
exerciseId (document ID)	string	false	Identity of the shared exercise definition; model id is populated from the document ID.
name	string	false	Displayed exercise name.
category	string	false	Exercise category, such as cardio, strength or mobility.
difficulty	string	false	Difficulty classification.
description	string	false	Exercise description and relevant guidance.
instructions	array(string)	false	Ordered instructional steps.
targetType	string	false	Target mode, such as duration or repetitions.
defaultTarget	integer	false	Default target amount for the exercise.
targetUnit	string	false	Target measurement unit, such as seconds.
equipment	array(string)	false	Required equipment identifiers.
metValue	number	false	MET parameter used for estimated energy expenditure.
imageUrl	string	true	Optional illustration URL; current catalogue entries may use an empty string.
active	boolean	false	Whether the exercise is available as active catalogue content.

A.8 WorkoutPaths collection

The workoutPaths collection organises guided lessons into named paths. Lesson content is stored in nested lessons documents rather than embedded in the path document.

Table A.8: WorkoutPaths collection

Field	Type	Nullable	Description
pathId (document ID)	string	false	Identity of the workout path.
name	string	false	Displayed path name.
description	string	false	Description of the path and its purpose.
active	boolean	false	Whether the path is active.

A.9 Lessons subcollection

Each lessons document defines a guided workout lesson, its progression position and eligibility information. The lesson's ordered exercise assignments are stored in a further subcollection.

Table A.9: Lessons subcollection

Field	Type	Nullable	Description
lessonId (document ID)	string	false	Identity of the lesson within the path.
name	string	false	Displayed lesson name.
description	string	false	Lesson description.
stageName	string	false	Displayed stage label.
sequence	integer	false	Ordering value within the path.
difficulty	string	false	Lesson difficulty.
estimatedMinutes	integer	false	Estimated workout duration in minutes.
baseXp	integer	false	Base XP used by guided reward calculations.
prerequisiteLessonId	string	true	Preceding lesson identifier; can be absent, null or empty for an entry lesson.
active	boolean	false	Whether the lesson is active.
standalone	boolean	false	Whether the lesson is a standalone alternative, such as recovery.
category	string	false	Category used by recommendation and weekly variety logic.
equipment	array(string)	false	Required equipment for eligibility filtering.

A.10 WorkoutExercises subcollection

The workoutExercises subcollection associates reusable exercise definitions with an ordered lesson. Its item identifier denotes the assignment, while exerciseId identifies the shared exercise definition.

Table A.10: WorkoutExercises subcollection

Field	Type	Nullable	Description
itemId (document ID)	string	false	Identity of the exercise assignment within a lesson.
exerciseId	string	false	Identifier of the corresponding exercises document.
sequence	integer	false	Exercise order within the lesson.
targetType	string	false	Duration or repetition target mode.
targetValue	integer	false	Required target amount for this lesson item.
restAfterSeconds	integer	false	Rest interval following the item.
optional	boolean	false	Indicates whether the item is optional.

A.11 WorkoutSessions subcollection

The workoutSessions subcollection records the immutable start context of a guided workout. The same session identifier is subsequently used for the completed workout record, supporting repeat-safe completion.

Table A.11: WorkoutSessions subcollection

Field	Type	Nullable	Description
sessionId (document ID)	string	false	Stable identifier for one guided session.
pathId	string	false	Selected workout path.
lessonId	string	false	Selected lesson.
plannedExercises	integer	false	Planned number of exercise items.
startedAt	timestamp	false	Server timestamp establishing the session start.

A.12 WorkoutRecords subcollection

The workoutRecords subcollection stores immutable guided-workout outcomes, submitted feedback and the reward result committed with each completion. It does not contain fabricated guided results for wearable-only activities.

Table A.12: WorkoutRecords subcollection

Field	Type	Nullable	Description
sessionId (document ID)	string	false	Matches the original workoutSessions identifier.
pathId	string	false	Completed workout path.
lessonId	string	false	Completed lesson.
lessonName	string	false	Historical lesson-name snapshot.
startedAt	timestamp	false	Start timestamp copied from the session document.
completedAt	timestamp	false	Server timestamp of completion processing.
plannedExercises	integer	false	Planned exercise-item count.
completedExercises	integer	false	Number of completed exercise items.
skippedExercises	integer	false	Number of skipped exercise items.
completionPercentage	number	false	Completed-item percentage relative to planned items.
durationSeconds	integer	false	Recorded session duration.
activeDurationSeconds	integer	false	Recorded active exercise duration.
estimatedCaloriesBurned	number	false	Estimated guided-workout energy expenditure.
xpEarned	integer	false	XP awarded by this completion event.
starsEarned	integer	false	Guided-result star count.
difficultyFeedback	string	false	User-reported difficulty: too_easy, suitable or too_difficult.
feelingFeedback	string	false	User-reported condition after exercise.

Field	Type	Nullable	Description
discomfortReported	boolean	false	Whether feeling feedback indicates discomfort.
firstCompletion	boolean	false	Whether this is the first saved attempt for the lesson.
totalXpAfter	integer	false	Account XP after the transaction.
weeklyCompleted	integer	false	Shared weekly count after processing.
weeklyTarget	integer	false	Weekly target used by the completion.
unlocked	array(string)	false	New achievement or lesson identifiers reported by the result.
localDay	string	false	Local date assigned to the completion.
localWeek	string	false	Monday-based local week identifier.
timeZone	string	false	Time-zone identifier used for date assignment.
category	string	false	Lesson category snapshot.

A.13 ExerciseLogs subcollection

The exerciseLogs subcollection stores dated manual activity entries and explicitly confirmed wearable sessions. Guided results remain in workoutRecords and are combined through dailyData accounting.

Table A.13: ExerciseLogs subcollection

Field	Type	Nullable	Description
date (document ID)	string	false	Local date of the activity log.
date	string	false	Date copied into the current log document.
exercises	array(map)	false	Manual or confirmed wearable activity entries.
totalCaloriesBurned	integer	false	Rounded known calories from this activity-log bucket only.
timestamp	timestamp	false	Latest activity-log update time.

Older aggregate-only documents may not contain an exercises array. The application preserves these through a legacy summary mapping rather than inventing exercise duration.

A.14 Exercises embedded activity records

An activity entry stores the amount and completion state of a manual activity or the provenance of a confirmed wearable session. Optional fields distinguish these two sources.

Table A.14: Exercises embedded activity records

Field	Type	Nullable	Description
entryId	string	false	Stable activity-entry identifier.
name	string	false	Activity name.
duration	number	false	Duration in minutes; wearable entries explicitly use elapsed duration.
calories	number	true	Estimated/manual or available wearable active calories; unavailable wearable energy can be null.
completed	boolean	false	Whether the entry contributes to counted totals.
revision	integer	false	Entry revision assigned by current mutation logic.
lastMutationId	string	true	Identifier supporting retry-safe manual edits.
source	string	true	manual for current manual entries or health_connect for confirmed wearable entries; older entries may omit it.
activityType	string	true	Supported manual-activity code.
metValue	number	true	MET value used by the manual estimate.
weightKg	number	true	Profile-weight snapshot used by the manual estimate.
calorieMethod	string	true	Manual estimation method, currently met_estimate.

Field	Type	Nullable	Description
legacySummary	boolean	true	Marks an aggregate-only historical summary or false for current manual entries.
sourcePackage	string	true	Source application package for wearable data.
healthConnectId	string	true	Original Health Connect session identifier.
startedAtMillis	integer	true	Wearable session start in epoch milliseconds.
endedAtMillis	integer	true	Wearable session end in epoch milliseconds.
calorieSource	string	true	wearable_active or unavailable for the wearable workflow.
durationSource	string	true	wearable_elapsed when derived from the wearable session interval.
workoutStartedAtMillis	integer	true	Time at which the user prepared the matching workout in FitnessBalance.
localDay	string	true	Recorded local date for a wearable entry.
timeZone	string	true	Time zone used to assign the wearable entry date.

Manual entries may include additional activity-specific estimation inputs. These embedded entries are not the same structure as the shared exercises catalogue.

A.15 SavedMeals subcollection

The savedMeals subcollection stores reusable groups of food-portion snapshots. Reuse creates new dated food-log entries while retaining the stored source and nutrition context.

Table A.15: SavedMeals subcollection

Field	Type	Nullable	Description
mealId (document ID)	string	false	Identity of the reusable meal.
name	string	false	User-defined meal name.

Field	Type	Nullable	Description
items	array(map)	false	One to 25 food snapshots, using the supported portion/nutrient/provenance fields from food entries.
revision	integer	false	Revision used to detect conflicting updates.
createdAt	timestamp	false	Creation timestamp.
updatedAt	timestamp	false	Latest update timestamp.

A.16 Recipes subcollection

The recipes subcollection stores ingredient snapshots and a recipe yield. Consumed recipe servings are calculated by scaling ingredient values relative to this yield.

Table A.16: Recipes subcollection

Field	Type	Nullable	Description
recipeId (document ID)	string	false	Recipe identity.
name	string	false	User-defined recipe name.
items	array(map)	false	Ingredient portion and nutrition snapshots.
yieldServings	number	false	Number of servings produced by the recipe.
revision	integer	false	Recipe revision.
mutationId	string	false	Identifier of the recipe save operation.
createdAt	timestamp	false	Creation timestamp.
updatedAt	timestamp	false	Latest update timestamp.

A.17 PersonalServings subcollection

The personalServings subcollection stores user-defined portion labels and measurable amounts for catalogue foods or private barcode products. The foodKey field identifies the applicable food without requiring duplicate shared catalogue documents.

Table A.17: PersonalServings subcollection

Field	Type	Nullable	Description
servingId (document ID)	string	false	Personal-serving identity.
name	string	false	User-defined serving label.

Field	Type	Nullable	Description
foodKey	string	false	Namespaced identity, such as food:<foodId> or barcode:<barcode>.
amount	number	false	Amount represented by the serving.
unit	string	false	Supported measurement unit, g or ml.
revision	integer	false	Serving revision used by update checks.
createdAt	timestamp	false	Creation timestamp.
updatedAt	timestamp	false	Latest update timestamp.

A.18 BarcodeFoods subcollection

The barcodeFoods subcollection stores private product definitions entered from package labels. Nutrients are normalised to a supported per-100-unit basis while retaining the original entered label information.

Table A.18: BarcodeFoods subcollection

Field	Type	Nullable	Description
barcode (document ID)	string	false	Validated product barcode used as the document identifier.
barcode	string	false	Product barcode copied into the record.
food_key	string	false	Private product key, generated with a user_barcode_ prefix.
food_name	string	false	User-entered product name.
food_name_lowercase	string	false	Normalised product name.
source_database	string	false	Description identifying the user-entered private product source.
nutrition_basis	string	false	Normalised per_100g or per_100ml basis.
portion_unit	string	false	g or ml.
nutrition	map	false	Available normalised canonical nutrient values.

Field	Type	Nullable	Description
label_amount	number	false	Reference amount transcribed from the package.
label_unit	string	false	Unit of the package reference amount.
label_nutrition	map	false	Original entered nutrients for the package reference amount.
nutrition_verification	string	false	user_entered provenance label.
basis_confirmation_required	boolean	false	False for the explicitly entered private label definition.

This documents the current UserBarcodeFood creation path. Shared external product lookup responses are not automatically equivalent to privately saved products.

A.19 FoodNutrition subcollection

The foodNutrition subcollection stores personal label-based nutrition or a user-confirmed approximate alternative for a shared food. These private overrides preserve the shared source record.

Table A.19: FoodNutrition subcollection

Field	Type	Nullable	Description
foodId (document ID)	string	false	Identity of the original shared food to which the override applies.
originalFoodId	string	false	Original food identifier.
food_name	string	false	Original/displayed food name.
nutrition	map	false	Supported private reference nutrient values.
nutrition_basis	string	false	Supported per_100g or per_100ml reference basis.
source_kind	string	false	label or alternative.
source_database	string	false	Description of the private nutrition source.

Field	Type	Nullable	Description
source_note	string	false	User/source note describing the information or approximate substitution.
nutrition_verification	string	false	Provenance label, such as user_entered or approximate_substitution.
label_amount	number	true	Original label reference amount for label-based entries.
label_unit	string	true	Original label measurement unit.
label_nutrition	map	true	Original values transcribed from the label.
alternativeFoodId	string	true	Selected alternative food identifier.
alternativeName	string	true	Selected alternative food name.
revision	integer	false	Override version.
mutationId	string	false	Save-operation identifier.
updatedAt	timestamp	false	Latest update timestamp.
confirmedAt	timestamp	false	Timestamp of explicit confirmation.

A.20 WeightLogs subcollection

The weightLogs subcollection stores one manual body-weight measurement per local date. Revision checks and an accompanying private index coordinate edits and removal.

Table A.20: WeightLogs subcollection

Field	Type	Nullable	Description
date (document ID)	string	false	Measurement date in yyyy-MM-dd format.
day	string	false	Measurement date copied into the document.
weightKg	number	false	Recorded weight in kilograms.
revision	integer	false	Measurement version.

Field	Type	Nullable	Description
source	string	false	manual for this workflow.
mutationId	string	false	Identifier of the save operation.
updateProfile	boolean	false	Whether saving this measurement also updates current profile weight.
updatedAt	timestamp	false	Latest update timestamp.

A.21 WeightHistory subcollection

The weightHistory subcollection maintains a private state index for weight-history operations. The entries map is updated atomically with the corresponding dated weight document.

Table A.21: WeightHistory subcollection

Field	Type	Nullable	Description
state (document ID)	string	false	Fixed index-document identifier.
entries	map	false	Date-keyed index; each value contains weightKg and revision.
updatedAt	timestamp	false	Latest index update timestamp.

This is a transaction-maintained index, not a second independent source of user-entered weight. The current implementation limits the index to 5,000 dates.

A.22 PathProgress subcollection

The pathProgress subcollection maintains guided lesson progression for each user and path. It tracks attempts and the best recorded result per lesson, supporting unlocking without altering shared lesson definitions.

Table A.22: PathProgress subcollection

Field	Type	Nullable	Description
pathId (document ID)	string	false	Shared workout path to which the progress belongs.
completionCounts	map	false	Lesson-ID keys mapped to saved attempt counts.
bestPercent	map	false	Lesson-ID keys mapped to best completion percentages.

Field	Type	Nullable	Description
bestStars	map	false	Lesson-ID keys mapped to best star counts.
updatedAt	timestamp	false	Latest progress update timestamp.

A.23 ExerciseWeeks subcollection

The exerciseWeeks subcollection stores shared weekly progress from eligible guided and confirmed wearable sessions. The document identifier is the Monday date of the corresponding local week.

Table A.23: ExerciseWeeks subcollection

Field	Type	Nullable	Description
week (document ID)	string	false	Monday-based week identifier in yyyy-MM-dd format.
completed	integer	false	Shared qualifying workout count for the week.
categories	array(string)	false	Recorded qualifying workout categories used by variety achievements.
bonusAwarded	boolean	false	Whether the weekly target bonus has already been awarded.
target	integer	false	Target used by the latest update.
lastRecordId	string	false	Guided record or wearable award responsible for the latest atomic update.
updatedAt	timestamp	false	Latest weekly-progress update timestamp.

A.24 Quests collection

The quests collection stores shared quest definitions. Completion evidence is stored separately in each user’s questProgress documents; editing a shared definition does not itself award progress.

Table A.24: Quests collection

Field	Type	Nullable	Description
questId (document ID)	string	false	Identity of the shared quest definition.
name	string	false	Displayed quest name.

Field	Type	Nullable	Description
target	integer	false	Target amount associated with the definition.
bonusXp	integer	false	Quest XP metadata.
active	boolean	false	Whether the definition is active in the seeded catalogue.

Some current quest labels and eligibility rules are defined in application resources/services. The catalogue document is not the sole executable source of reward policy.

A.25 QuestProgress subcollection

The questProgress subcollection records immutable daily quest completions. A date-plus-quest identifier allows guided and wearable reward paths to share the same daily completion receipt.

Table A.25: QuestProgress subcollection

Field	Type	Nullable	Description
date_questId (document ID)	string	false	Composite date and quest identifier.
day	string	false	Local date to which completion applies.
questId	string	false	Completed quest identifier.
completed	boolean	false	True for a committed completion.
xpEarned	integer	false	Quest-specific XP recorded for the completion.
recordId	string	false	Guided record or wearable award supporting the completion.
completedAt	timestamp	false	Server timestamp of progress creation.

A.26 Achievements collection

The achievements collection stores shared achievement display metadata. Eligibility is evaluated by the application’s guided and wearable reward services and constrained by the relevant Firestore rules.

Table A.26: Achievements collection

Field	Type	Nullable	Description
achievementId (document ID)	string	false	Identity of the achievement definition.
name	string	false	Displayed achievement name.

Field	Type	Nullable	Description
description	string	false	Description of the achievement.
active	boolean	false	Whether the seeded definition is active.

A.27 UnlockedAchievements subcollection

The unlockedAchievements subcollection stores the user's immutable achievement-unlock evidence. Reusing the shared achievement identifier prevents the same achievement from being created repeatedly for that user.

Table A.27: UnlockedAchievements subcollection

Field	Type	Nullable	Description
achievementId (document ID)	string	false	Identity of the shared achievement that was unlocked.
recordId	string	false	Guided record or wearable award that produced the unlock.
unlockedAt	timestamp	false	Server timestamp of the unlock.

A.28 SavedMealLogs subcollection

The savedMealLogs subcollection stores operation receipts for inserting a saved meal or recipe into the food diary. A repeated operation identifier returns an existing result rather than appending its foods again.

Table A.28: SavedMealLogs subcollection

Field	Type	Nullable	Description
operationId (document ID)	string	false	UUID identifying the reusable-food insertion operation.
date	string	false	Destination food-log date.
mealId	string	false	Source reusable meal identity; recipe reuse can use a recipe-prefixed identity.
entryCount	integer	false	Number of food entries inserted by the operation.
createdAt	timestamp	false	Server timestamp of the receipt.

A.29 WearableWorkoutLinks subcollection

The wearableWorkoutLinks subcollection stores immutable confirmation receipts for sessions explicitly saved from Health Connect. Its stable identifier is derived from the source package and original session ID.

Table A.29: WearableWorkoutLinks subcollection

Field	Type	Nullable	Description
entryId (document ID)	string	false	Stable wearable_ key followed by the SHA-256 digest of source/session identity.
date	string	false	Date of the corresponding confirmed exercise-log entry.
entryId	string	false	Stable activity identifier copied into the receipt.
createdAt	timestamp	false	Server timestamp of the confirmed insertion.

Removing the activity entry does not remove this receipt. This prevents accidental re-import of the same session after deletion. Syncing a local wearable cache alone does not create this document.

A.30 WearableWorkoutRewards subcollection

The wearableWorkoutRewards subcollection stores immutable, versioned reward outcomes for qualifying confirmed wearable sessions. It is separate from guided workoutRecords so that a wearable activity does not imply completion of a guided lesson.

Table A.30: WearableWorkoutRewards subcollection

Field	Type	Nullable	Description
entryId (document ID)	string	false	Stable identifier of the confirmed wearable activity.
entryId	string	false	Activity identifier copied into the award.
source	string	false	health_connect.
activity	string	false	Activity-name snapshot.
day	string	false	Workout date receiving credit.
week	string	false	Monday-based event week receiving credit.

Field	Type	Nullable	Description
category	string	false	Supported reward category assigned to the activity.
startedAtMillis	integer	false	Original session start in epoch milliseconds.
endedAtMillis	integer	false	Original session end in epoch milliseconds.
durationSeconds	integer	false	Elapsed duration used by reward eligibility.
baseXp	integer	false	Base wearable XP, capped by the date credit receipt.
questXp	integer	false	Previously unclaimed shared daily quest bonuses.
weeklyBonusXp	integer	false	Previously unclaimed shared weekly target bonus.
xpEarned	integer	false	Sum of the XP components for this event.
weeklyCompleted	integer	false	Shared week count after the award.
weeklyTarget	integer	false	Target used in the reward decision.
totalXpAfter	integer	false	Account XP after the award transaction.
newDayCredit	boolean	false	Whether this event received the date's wearable base-XP and weekly credit.
unlocked	array(string)	false	General achievement identifiers newly unlocked by this event.
ruleVersion	string	false	Version of the wearable reward policy.
createdAt	timestamp	false	Server timestamp of award creation.

The current policy requires a recent confirmed session of at least five elapsed minutes. Calories do not determine XP. Display-only fields such as eligible or alreadyAwarded are transient results, not stored ledger fields.

A.31 WearableRewardDays subcollection

The wearableRewardDays subcollection stores one immutable wearable credit receipt per workout date. It coordinates simultaneous devices and limits wearable base XP and weekly credit independently of how many sessions are imported.

Table A.31: WearableRewardDays subcollection

Field	Type	Nullable	Description
date (document ID)	string	false	Workout date receiving the wearable credit.
day	string	false	Date copied into the document.
week	string	false	Event week associated with that date.
awardId	string	false	Wearable award that first claimed the date credit.
createdAt	timestamp	false	Server timestamp of credit creation.

A.32 Local Device Storage

The following local structures support offline state, interrupted-workout recovery, permission-aware health display and optional recommendation functions. They are separate from Firestore. Explicitly confirmed wearable activity and eligible reward evidence can be stored in the cloud, but synchronisation alone does not upload the full local health cache.

Table A.32: Local device storage structures

Structure	Storage mechanism	Purpose
Health summaries	HealthSummaryStore private no-backup JSON	Permission/freshness state and supported steps, sleep and heart-rate summaries.
Wearable sessions and pending preparation	WearableWorkoutStore private no-backup SQLite	Rolling imported-session cache, sync metadata and account-specific pending workout.
Guided workout draft	WorkoutDraftStore private SharedPreferences	Resume state before immutable guided completion is submitted.
Sleep recommendation context	HealthRecommendationStore local cache	Opt-in sleep comparison baseline and bounded explanation history.
Optional contextual-bandit state	BanditStore local state	Local decision context, model parameters and processed-outcome evidence.
Food recommendation explanation	Account-private on-device snapshot	Latest rule inputs and explanation context.

Structure	Storage mechanism	Purpose
Celebration display receipts	RewardCelebrationStore private no-backup SQLite	Prevents repeated guided celebration display; does not authorise rewards.

A.33 Logical Collection Relationships

The relationships below are established through parent paths, document identifiers and references held in documents or embedded records. They are logical associations rather than SQL foreign-key constraints. Source snapshots and receipts can retain identifiers after their originating definitions are removed.

Table A.33: Logical collection relationships

Relationship	Implementation	Cardinality / qualification
Authentication identity → users	Firebase Authentication UID matches users document ID.	One identity can have one application profile.
users → private subcollections	Ownership is established by users/{uid}.	One user can own many dated or event records; some state documents are singleton.
workoutPaths → lessons	Parent/subcollection nesting.	One path contains multiple lessons.
lessons → workoutExercises	Parent/subcollection nesting.	One lesson contains multiple ordered assignments.
exercises → workoutExercises	exerciseId points to a reusable definition.	One exercise can be assigned to multiple lesson items.
workoutSessions → workoutRecords	The same session ID is reused for the result.	A started session has zero or one immutable completion.
Food definitions → foodLogs	foodItems stores source IDs and historical values.	Many-to-many at dated-document level through embedded snapshots.
savedMeals / recipes → foodLogs	New snapshots retain source and operation identifiers.	One reusable definition can supply multiple diary insertions.
Source logs → dailyData	Related source operations update the same user/date summary.	Derived summary dependency, not an enforced foreign key.
weightLogs → weightHistory	Dated record and index entry are updated together.	Multiple dates are represented in at most one private state index per user.
workoutPaths → pathProgress	Progress document ID identifies the shared path.	At most one progress document per user and path.
quests → questProgress	questId and date form the daily completion identity.	A user can have one completion per date and quest.
achievements → unlockedAchievements	The shared achievement ID is the private document ID.	At most one immutable unlock per user and achievement.

Relationship	Implementation	Cardinality / qualification
Confirmed wearable entry → wearableWorkoutLinks	Stable entry ID derives from source package and Health Connect ID.	One confirmation receipt per source session; it can outlive deletion of the log entry.
wearableWorkoutLinks → wearableWorkoutRewards	The stable confirmed entry ID identifies the award.	A qualifying confirmed session has at most one immutable award.
wearableWorkoutRewards → wearableRewardDays	awardId identifies the first qualifying wearable award for the date.	At most one date-credit receipt per user and workout date.
Guided / wearable award → shared progress	recordId can name either event source.	The sources share week, quest and profile progress without creating fictitious guided results.